

Kubernetes Assignment

Step-by-Step Implementation Report

1. Minikube Images Verification

Verified Docker images inside Minikube (v1 & v2).

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> minikube image ls
registry.k8s.io/pause:3.10.1
registry.k8s.io/kube-scheduler:v1.35.1
registry.k8s.io/kube-proxy:v1.35.1
registry.k8s.io/kube-controller-manager:v1.35.1
registry.k8s.io/kube-apiserver:v1.35.1
registry.k8s.io/etcd:3.6.6-0
registry.k8s.io/coredns/coredns:v1.13.1
gcr.io/k8s-minikube/storage-provisioner:v5
docker.io/library/k8s-app:v2
docker.io/library/k8s-app:v1
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> █
```

2. Pod Creation

Created single pod and verified running state.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl apply -f k8s/pod.yml
pod/demo-pod created
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
demo-app-deployment-58848fd6f9-26p6v	1/1	Running	0	6m10s
demo-app-deployment-58848fd6f9-bd9jt	1/1	Running	0	6m11s
demo-app-deployment-58848fd6f9-q6x2z	1/1	Running	0	6m8s
demo-app-deployment-probe-c99c65ccc-4xlpv	1/1	Running	0	4m52s
demo-app-deployment-probe-c99c65ccc-6mjq6	1/1	Running	0	4m51s
demo-app-deployment-probe-c99c65ccc-c2zqq	1/1	Running	0	4m51s

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```

3. ReplicaSet

ReplicaSet created multiple pods ensuring scaling.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl apply -f k8s/rs.yml
replicaset.apps/demo-rs configured
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
demo-app-deployment-58848fd6f9-26p6v	1/1	Running	0	8m
demo-app-deployment-58848fd6f9-bd9jt	1/1	Running	0	8m1s
demo-app-deployment-58848fd6f9-q6x2z	1/1	Running	0	7m58s
demo-app-deployment-probe-c99c65ccc-4xlpv	1/1	Running	0	6m42s
demo-app-deployment-probe-c99c65ccc-6mjq6	1/1	Running	0	6m41s
demo-app-deployment-probe-c99c65ccc-c2zqq	1/1	Running	0	6m41s

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```

4. Deployment

Deployment created with 3 replicas.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl apply -f k8s/deployment.yml
deployment.apps/demo-app-deployment configured
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```

5. Service

NodePort service exposed application.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
demo-service	NodePort	10.100.185.253	<none>	80:30007/TCP	13m
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	37m

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```

6. Version 1 Output

Application showing Version 1.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> minikube service demo-service
```

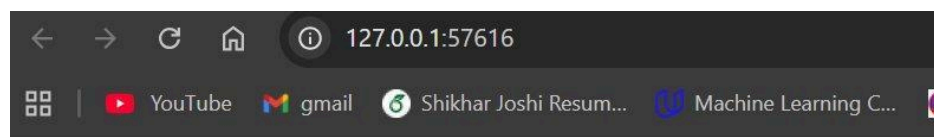
NAMESPACE	NAME	TARGET PORT	URL
default	demo-service	80	http://192.168.49.2:30007

🔑 Starting tunnel for service demo-service.

NAMESPACE	NAME	TARGET PORT	URL
default	demo-service		http://127.0.0.1:57616

🌐 Opening service default/demo-service in default browser...

! Because you are using a Docker driver on windows, the terminal needs to be open to run it.



Version 1

7. Rolling Update

Updated image to v2.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl apply -f k8s/deployment.yml
deployment.apps/demo-app-deployment configured
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```

8. Rolling Update Progress

Pods replaced gradually.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl get pods -w
```

NAME	READY	STATUS	RESTARTS	AGE
demo-app-deployment-6d6987f4df-2w8jn	1/1	Running	0	4m3s
demo-app-deployment-6d6987f4df-4kzkd	1/1	Running	0	4m1s
demo-app-deployment-6d6987f4df-zn54t	1/1	Running	0	4m2s
demo-app-deployment-probe-c99c65ccc-4xlpv	1/1	Running	0	11m
demo-app-deployment-probe-c99c65ccc-6mjg6	1/1	Running	0	11m
demo-app-deployment-probe-c99c65ccc-c2zqq	1/1	Running	0	11m

9. Version 2 Output

Application showing Version 2.



Version 2

10. Probes Deployment

Deployment with liveness & readiness probes.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl apply -f k8s/deployment_probe.yml
deployment.apps/demo-app-deployment-probe configured
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```

11. Probe Pods

Pods running with probes.

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
demo-app-deployment-6d6987f4df-2w8jn 1/1     Running   0           4m3s
demo-app-deployment-6d6987f4df-4kzkd 1/1     Running   0           4m1s
demo-app-deployment-6d6987f4df-zn54t 1/1     Running   0           4m2s
demo-app-deployment-probe-c99c65ccc-4xlpv 1/1     Running   0           11m
demo-app-deployment-probe-c99c65ccc-6mjqq 1/1     Running   0           11m
demo-app-deployment-probe-c99c65ccc-c2zqq 1/1     Running   0           11m
```

12. Rollout History

Deployment revision history (v1 → v2).

```
PS C:\AI-ML-ASSIGNMENTS\Assignment-3> kubectl rollout history deployment demo-app-deployment
deployment.apps/demo-app-deployment
REVISION  CHANGE-CAUSE
0          <none>
2          Updated to v2
3          Initial deployment v1
PS C:\AI-ML-ASSIGNMENTS\Assignment-3>
```